Earth–Mars Mission Mathematics and Simulation Evidence

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To the engineering teams at SpaceX and NASA:

This study sets out a reproducible route from Earth departure conditions to a Mars arrival and a physically constrained landing model. It supplies the equations, data contracts, calculated reference cases, and simulation evidence needed to examine such a route. It keeps three different things explicit: an analytic comparison, a dated trajectory computed from NASA/JPL observations-based ephemerides, and a separately identified application-run evidence section for the experimental EX-1 vehicle.

A trajectory that reaches Mars is only part of a landing design. An operational solution also needs the actual vehicle's aerodynamic and propulsion models, landing-site terrain, atmospheric dispersions, navigation covariance, and verified flight constraints. There is no epoch-independent angle or fuel quantity that lands every Starship configuration. All numbers below identify their frame, units, epoch, architecture, and assumptions.

1. Deliverables and reproduction

The numerical companion is earth_mars_reference.py. The complete precision output is earth_mars_reference_results.json. Exact JPL response files and their provenance are in earth_mars_data. The printable HTML and PDF are generated from this Markdown source and the telemetry attachments; they do not depend on a remote mathematics renderer.

```
# Offline: use the preserved JPL responses; reproduce both worked examples.
python3 Sandbox/Space/reports/earth_mars_reference.py

# Optional explicit refresh: retrieve two small, read-only JPL vector tables.
python3 Sandbox/Space/reports/earth_mars_reference.py --fetch-horizons

# Render the report; add the complete application run when available.
python3 Sandbox/Space/reports/render_earth_mars_study.py \
  --telemetry Sandbox/Space/reports/telemetry/earth_mars_run.json \
  --telemetry-csv Sandbox/Space/reports/telemetry/earth_mars_run.csv
```

The calculator uses Python's standard library. The renderer uses Python-Markdown and ReportLab; its output manifest records their installed versions and all input hashes. All telemetry rows stay in the attached JSON and CSV rather than being reduced to selected points in the document. Document plots may downsample the displayed line; the original attached records are retained.

2. Data required before selecting a mission

Domain	Required inputs	Why they matter
Time and geometry	Departure/arrival epochs, time scale, ephemeris version, reference axes and origin	The same numerical coordinates mean different things in different frames or at different epochs.
Earth departure	Launch site, Earth orientation, launch azimuth constraints, parking orbit, vehicle state and covariance	A heliocentric transfer direction does not specify a surface launch bearing.
Vehicle	Dry/payload/propellant masses, center of mass and inertia, tank geometry, usable residuals, thrust and Isp maps, throttle limits, engine-out rules	Mass, burn duration, attitude authority, and fuel reserve depend on the selected configuration.
Cruise	Power/thermal pointing, communications, radiation environment, navigation measurements, correction allocation, boiloff and transfer losses	A passive two-body coast does not model all operational constraints.
Mars arrival	Target B-plane, entry interface, landing site and season, atmosphere/wind distributions, gravity field, terrain and hazards	Entry state must fall inside a vehicle-specific corridor and reachable terrain region.
Entry/descent	Aerodynamic coefficient tables, TPS response, plume interaction, control surfaces, sensor performance, propulsion transients	Neither a generic drag coefficient nor a fixed landing angle can establish landing feasibility.
Acceptance	Dispersions, probability requirements, reserve policy, structural/thermal limits, navigation and guidance tolerances	A nominal trajectory passing through a point is insufficient evidence.

NASA's Mars-GRAM 2024 estimates atmospheric mean conditions and statistical variations. It supplies an engineering environment, not one universal atmosphere for every place and time. A mission should preserve the exact configuration, input data, and random seeds used for its dispersion study. NASA Mars-GRAM 2024 user guide.

3. Frames, units, time, and the moving Solar System

Use SI units for dynamics and mass accounting, or an explicitly consistent kilometre-second unit system for orbital calculations. Here the orbital calculator uses km, km/s and km³/s², then explicitly converts delta-v to m/s for the rocket equation. Angles are radians inside algorithms and degrees in tables. A tonne is 1,000 kg. Conventional standard acceleration is `g0 = 9.80665 m/s²`; it is not local Mars gravity.

The dated example uses geometric states in ICRF axes, with the Sun as origin, and TDB epochs. Earth is target 399 and Mars is target 499; these are body centers rather than Earth–Moon or Mars-system barycenter target IDs. Horizons requests explicitly select `VECTORS`, `REF_SYSTEM=ICRF`, `REF_PLANE=FRAME`, `CENTER=500@10`, `VEC_CORR=NONE`, `OUT_UNITS=KM-S`, and `TIME_TYPE=TDB`. Exact requests and response hashes accompany the data. JPL Horizons API documentation.

Use geometric simultaneous states for force integration and targeting. A sky display may instead need light-time and stellar-aberration corrections; those observed directions cannot be substituted as simultaneous force-model positions. In SPICE, `spkezr` returns state vectors with an explicit frame, observer and aberration setting. NAIF state-vector interface.

Civil UTC, TT, TDB and UT1 serve different purposes. Convert UTC to the integration time scale using a versioned leap-second kernel and an appropriate time model; use Earth-orientation data for Earth-fixed transformations. The companion's date parser intentionally treats the supplied calendar labels as TDB and does not silently claim to convert UTC. NAIF time subsystem.

For an inertial barycentric formulation, let `r_i` be each body's position and `mu_j = G m_j`:

```
dr_i/dt = v_i
dv_i/dt = sum_(j != i) mu_j (r_j - r_i) / |r_j - r_i|^3 + a_non_gravitational
```

The Sun responds to other bodies; it need not be fixed at an artificial origin. A common velocity added to every object changes inertial positions but preserves relative velocities and Newtonian encounters. In particular, a large Galactic translation must not be included in an Earth departure's `v_infinity` or the ship's heliocentric attitude heading.

In a Sun-centered accelerating frame, third-body accelerations need the indirect term:

```
a_third_body,j = mu_j [ (r_j-r_sc)/|r_j-r_sc|^3 - r_j/|r_j|^3 ]
```

An external Galactic potential mainly supplies common Solar System acceleration on an interplanetary mission. Its differential tidal field is the relevant relative effect. A Galaxy viewer's catalogue objects are not automatically a complete self-consistent mass model: missing stellar masses, gas and dark matter must not be silently replaced by equal-mass points. Do not double-count a star as both a resolved gravitating body and part of an unresolved smooth potential.

For precise mission work, use versioned observationally fitted ephemerides rather than describing a Kepler approximation as a precision ephemeris. JPL's DE440/DE441 work combines numerical dynamics with fitted observations; differing time-span and modeling choices matter. JPL DE440/DE441 publication.

4. A force-model hierarchy that can be validated

The following is a mission-model specification, not a claim that every term is implemented by the application or the analytic companion.

Phase	Baseline dynamics	Refinements to assess and validate
Broad window search	Sun-centered two-body Lambert transfers	Multiple branches, departure/arrival constraints, launch energy, inclination and entry-state bounds
Committed interplanetary trajectory	Consistent planetary ephemerides, finite burns, navigation corrections	Planet/Moon perturbations, solar radiation pressure, thermal recoil if material, relativistic terms at the required accuracy
Earth departure / Mars orbit	Body-centered numerical propagation	Nonspherical gravity, rotation, third bodies, drag and eclipse geometry where relevant
Entry/descent	Six-degree-of-freedom vehicle and mass dynamics	Atmospheric dispersions, gas chemistry/heating, aero tables, thrust/plume interaction, actuator and sensor dynamics
Touchdown	Terrain-relative motion and contact dynamics	Landing gear compliance, friction, slopes, soil response, plume–surface interaction and stability

For example, a solar test-particle first post-Newtonian correction in a suitable Sun-centered approximation is:

```
a_1PN = mu_sun/(c² r³) [ (4 mu_sun/r - v²) r_vector
                         + 4 (r_vector dot v_vector) v_vector ]
```

This compact term is not the complete relativistic N-body model. It must not be blindly added to an already supplied spacecraft ephemeris. The integration model, ephemeris definition and observable corrections must be consistent. An accuracy target and convergence/error budget are more useful than attaching a new model name without validation.

Near an oblate body's polar axis frame, a common `J2` acceleration added to the central point-mass term is:

```
K = 3 J2 mu R² / (2 r^5)
a_J2 = K [ x(5z²/r²-1), y(5z²/r²-1), z(5z²/r²-3) ]
```

The gravity coefficients, normalization and body orientation must match the source model. Solar radiation pressure can start with a shadow-aware cannonball approximation, `a_SRP = P_1AU C_R (A/m) (1 AU/r)² s_hat`, where `s_hat` points

away from the Sun. Coefficient, area, attitude and shadow assumptions should remain visible in metadata.

Convergence must be demonstrated by step/tolerance reduction and independent propagation. Conservative-force validation includes energy and angular-momentum residuals; finite thrust and atmospheric flight require accounting for work, mass flow and dissipative terms rather than demanding spurious energy conservation. The study companion checks its Lambert solution with a separate Cartesian RK4 integration; agreement is a numerical consistency test, not real-flight validation.

5. Analytic reference: circular, coplanar Hohmann transfer

Assume a circular Earth orbit of `r_E = 1 AU`, a circular Mars orbit of `r_M = 1.52371034 AU`, and a Sun point mass. The Mars radius in this ideal comparison uses a nominal J2000 semimajor-axis value; eccentricity, actual orbital phases and inclination are deliberately omitted. The common transfer plane is an arbitrary ideal plane, not Earth's equator. The model is useful for dimensional checks and intuition. JPL approximate planetary elements, NASA trajectory fundamentals.

```
a_T = (r_E + r_M)/2
v_circular(r) = sqrt(mu_sun/r)
v_transfer(r) = sqrt(mu_sun [2/r - 1/a_T])
t_flight = pi sqrt(a_T³/mu_sun)
n_E = sqrt(mu_sun/r_E³),  n_M = sqrt(mu_sun/r_M³)
P_synodic = 2 pi / |n_E - n_M|
Mars lead at departure = pi - n_M t_flight
```

Calculated quantity	Reference value
Transfer semimajor axis	1.26185517 AU
Time of flight	258.870581 days
Ideal synodic recurrence	779.922127 days
Mars ahead of Earth at departure, measured around the Sun	44.345829°
Earth's heliocentric speed	29.784692 km/s
Transfer speed at departure	32.729514 km/s
Outbound Earth excess speed	2.944823 km/s
Mars heliocentric speed at arrival	24.129143 km/s
Transfer speed at arrival	21.480142 km/s
Mars arrival excess speed magnitude	2.649001 km/s

The phase angle is a Sun-centered planet-to-planet angle at departure. It is not a rocket pitch angle, launch-site bearing or fixed latitude. The recurrence above is an ideal orbital-period calculation, not a schedule of approved launch opportunities.

The constants used are `AU = 149597870.7 km`, `mu_sun = 1.32712440041279419e11 km³/s²`, `mu_Earth = 398600.435507 km³/s²`, and `mu_Mars-system = 42828.375816 km³/s²`. The reference approximates the Mars-system parameter as a central point mass; precision local Mars modeling must distinguish Mars from its moons. Reference spherical surface radii are Earth's equatorial `6378.1366 km` and Mars's equatorial `3396.19 km`. JPL astrodynamic parameters, JPL physical parameters.

6. Earth departure: excess velocity, parking orbit and the actual burn direction

The departure excess-velocity vector is the difference between the spacecraft's heliocentric transfer velocity and Earth's heliocentric velocity:

```
v_inf,E = v_sc,departure - v_Earth,departure
C3 = |v_inf,E|²
r_p = R_Earth + parking_altitude
v_p,hyp = sqrt(|v_inf,E|² + 2 mu_Earth/r_p)
v_p,circ = sqrt(mu_Earth/r_p)
Delta-v_TMI = v_p,hyp - v_p,circ             [aligned impulsive case]
```

For the ideal case from a 200 km circular orbit, `r_p = 6578.1366 km`, `C3 = 8.671980 km²/s²`, the circular speed is `7.784262 km/s`, hyperbolic periapsis speed is `11.395677 km/s`, and the aligned impulsive injection is `3.611415 km/s`. This excludes Earth ascent, finite-burn losses and plane mismatch.

Hyperbola geometry determines the periapsis orientation relative to the outgoing asymptote:

```
e_h = 1 + r_p |v_inf|²/mu_Earth
nu_inf = acos(-1/e_h)
total flyby deflection = 2 asin(1/e_h)
b = r_p sqrt(1 + 2 mu_Earth/[r_p |v_inf|²])
```

Here `e_h = 1.143114`, `nu_inf = 151.021677°`, and the complete incoming-to-outgoing deflection would be `122.043354°`. These are different angles and must not be interchanged.

For a reproducible arbitrary-plane construction, place Earth at heliocentric `(+r_E,0,0)`, moving along `+Y`, with prograde normal `+Z`. The ideal outgoing Earth excess velocity is also `+Y`. One departure periapsis radius points at `-61.021677°` from `+X`; the tangential injection direction points at `+28.978323°` from `+X`. The burn is locally horizontal/prograde in the assumed circular parking orbit. These two angles are illustrative coordinate components, not Earth-fixed launch commands.

In the general 3D case, let `s_hat = unit(v_inf,E)` and select an admissible escape-plane normal `n_hat` with `n_hat dot s_hat = 0`. Construct:

```
p_hat = rotate_about(n_hat, -nu_inf) s_hat
q_hat = n_hat cross p_hat
r_periapsis = r_p p_hat
v_periapsis = v_p,hyp q_hat
Delta-v_vector = v_periapsis - v_actual_parking_orbit
```

The launch location, parking orbit and timing constrain which `n_hat` is reachable. If inertial asymptote declination is `delta`, a necessary geometric condition is an orbit-plane inclination at least `|delta|`; launch latitude and operational azimuth restrictions add further constraints. An orbit plane and a departure asymptote do not determine a surface launch command by themselves.

For a nonrotating spherical Earth and a directly reached prograde orbit, the familiar launch-bearing relation is `cos(i) = cos(latitude) sin(azimuth)`, with azimuth measured clockwise from north. Actual ascent targeting must include Earth's rotation, winds, finite thrust, flight-path shaping and range constraints. A plane change costs `2 v sin(Delta-i/2)` for equal-speed impulsive reorientation; combining speed and plane changes gives `sqrt(v1² + v2² - 2 v1 v2 cos(Delta-i))`.

7. Dated 3D reference from preserved NASA/JPL vectors

This example selects two explicit TDB epochs to demonstrate the full calculation. It is not an optimized or recommended mission: departure is **20 November 2026, 00:00:00 TDB**, arrival is **18 July 2027, 00:00:00 TDB**, and time of flight is **240 days**. The state-vector snapshots preserve the exact JPL output used.

Lambert's boundary-value problem is:

```
Find v1 and v2 such that a trajectory under -mu_sun r/|r|³ has:
r(t_departure) = r_Earth(t_departure)
r(t_arrival)   = r_Mars(t_arrival)
t_arrival - t_departure = selected flight time
```

The companion solves a zero-revolution universal-variable branch with the prograde sense selected relative to Earth's instantaneous orbital normal. It rejects unsupported collinear/singular cases. Operational window searches must compare all admissible branches and refine the selected finite-burn trajectory. A Lambert result is a seed for further targeting, not proof of landing feasibility. NASA trajectory survey method, original Lambert algorithm research.

Using Stumpff functions `C(z)` and `S(z)`, one convenient formulation is:

```
A = sin(Delta-theta) sqrt(r1 r2/[1-cos(Delta-theta)])
y(z) = r1 + r2 + A [z S(z)-1]/sqrt(C(z))
t(z) = { [y/C]^(3/2) S(z) + A sqrt(y) } / sqrt(mu_sun)
Solve t(z) = selected time of flight
f = 1-y/r1,  g = A sqrt(y/mu_sun),  g_dot = 1-y/r2
v1 = (r2_vector - f r1_vector)/g
v2 = (g_dot r2_vector - r1_vector)/g
```

Dated-case result	Value, in ICRF axes unless noted
Departure excess velocity vector	(-1.500051, +2.778535, +1.920817) km/s
Departure excess speed	3.695937 km/s
Departure `C3`	13.659947 km²/s²
Departure asymptote right ascension / declination	118.363344° / +31.312835°
Arrival excess velocity vector	(-3.496342, +0.180817, +0.473340) km/s
Arrival excess speed	3.532867 km/s
Arrival velocity-vector right ascension / declination	177.039531° / +7.699741°
Injection from an aligned 200 km circular Earth orbit	3.828206 km/s
Capture into an aligned 250 km circular Mars orbit	2.570520 km/s
One illustrative escape-plane inclination to the ICRF equator	33.023688°
Its unit injection direction	(+0.508812, +0.743325, +0.434257)
Its injection-direction right ascension / declination	55.608056° / +25.738035°
Mars–Earth instantaneous center distance at arrival	252,171,731.753 km
Instantaneous distance divided by light speed	841.154 seconds, approximately 14.019 minutes

The displayed arrival vector points along the planet-relative incoming velocity. An arrival radiant points in the opposite direction. The one-way light-time estimate above is instantaneous `range/c`; a precise link observable solves retarded

transmission/reception geometry and adds the propagation corrections appropriate to the measurement.

The complete heliocentric boundary states, transfer velocities, hyperbola parameters, escape-plane construction and independent endpoint residual are in the JSON companion. No out-of-plane component is projected away. An independent 1,800 s-step Cartesian RK4 propagation arrives within approximately `4.05e-6 km` of the Lambert endpoint for this case. That small mathematical residual does not represent actual trajectory uncertainty or landing accuracy.

8. Propellant accounting, reserves and finite burns

For a constant equivalent exhaust velocity, the ideal rocket equation is:

```
v_e = g0 Isp
Delta-v = v_e ln(m_before/m_after)
m_before = m_after exp(Delta-v/v_e)
m_propellant = m_before - m_after
dm/dt = -T/(g0 Isp)
```

Propellant includes fuel and oxidizer. For oxidizer/fuel mass ratio `OF`, `m_fuel = m_propellant/(1+OF)` and `m_oxidizer = OF m_fuel`. An Isp value belongs to an operating condition and engine configuration, not to the Mars atmosphere or a universal vessel. NASA Glenn specific impulse, NASA rocket-equation discussion.

For sequential burns, work backward from the retained final mass. Each earlier burn must accelerate all propellant retained for later burns. Simply adding fuel estimates calculated with the same starting mass gives the wrong result. Include hardware jettison, transfers, leakage, unusable residuals and uncertainty explicitly between burn segments.

The following is an **ideal arithmetic ledger**, not a Starship mission fuel budget. It uses the analytic Hohmann injection, assumes a 100 m/s correction allocation, assumes aerodynamic deceleration can deliver the terminal reference state in Section 10, assumes 380 s cruise Isp and 330 s terminal Isp, and retains 200,000 kg at touchdown. Those vehicle/performance values are user-adjustable study assumptions; no public SpaceX performance map was used to validate them.

Step	Mass before, kg	Propellant consumed, kg	Mass after, kg
Earth orbital injection, 3,611.415 m/s at assumed 380 s	571,411.980	354,606.531	216,805.448
Assumed corrections, 100 m/s at assumed 380 s	216,805.448	5,740.529	211,064.920
Ideal final braking, 174.264 m/s at assumed 330 s	211,064.920	11,064.920	200,000.000
Total	571,411.980	371,411.980	200,000.000

If one additionally assumes `OF = 3.6`, the ledger's total consists of about `80,741.735 kg` fuel and `290,670.245 kg` oxidizer. This mixture ratio is an illustrative input, not a certified Raptor value. The ledger excludes Earth ascent, orbital refilling losses, boiloff, finite-burn losses, reserve policy, hover/divert, plane changes, entry thrust, surface stay and Earth return. The unvalidated aerodynamic step is particularly consequential.

A propulsive-capture architecture must additionally budget capture, deorbit and the changed entry state; the 250 km circular capture cost cannot be silently omitted or simultaneously credited as free atmospheric braking. Conversely, direct entry does not require paying a circularization burn first. These are alternative mission architectures, not interchangeable labels for the same burn sequence.

For a finite burn, integrate `T(t)`, `Isp(t)`, mass, position and orientation. A simple `burn_time = m_propellant g0 Isp/T` is valid only for constant thrust/Isp and the specified consumed mass. Long Earth-periapsis burns cannot generally be replaced by one instantaneous impulse without assessing losses, arc splitting and retargeting. Capacity alone does not establish delivered delta-v.

9. Mars approach, capture or entry, and B-plane targeting

For the analytic excess speed of `2.649001 km/s`, a point-mass 250 km Mars capture has incoming periapsis speed `5.523525 km/s` and circular speed `3.427253 km/s`, yielding an aligned impulsive capture cost of `2.096273 km/s`. The dated example has different values because its boundary geometry is different.

For an arbitrary target ellipse with periapsis `r_p` and apoapsis `r_a`:

```
a_target = (r_p+r_a)/2
v_target,p = sqrt(mu_Mars [2/r_p - 1/a_target])
Delta-v_capture = |v_incoming,p_vector - v_target,p_vector|
```

Capture must be targeted at the correct position and orientation, not just the correct radius. In a B-plane construction, choose the incoming asymptote convention explicitly. For example, use `S_hat = unit(v_inf,in)`, pick a nonparallel reference axis, define `T_hat = unit(reference cross S_hat)`, and `R_hat = S_hat cross T_hat`. The B-plane coordinates `B_T` and `B_R` specify the transverse offset. Different software packages use different signs/reference axes; exchange the basis vectors alongside the numbers.

In an unpowered point-mass hyperbola, impact parameter magnitude and periapsis are related by:

```
b = (mu_Mars/v_inf²) sqrt(e_h²-1)
e_h = 1 + r_p v_inf²/mu_Mars
```

At a selected atmospheric interface radius `r_EI`, define inertial flight-path angle by `gamma = asin(r_hat dot v_hat)`; descending is negative. Before appreciable drag:

```
v_EI = sqrt(v_inf² + 2 mu_Mars/r_EI)
h_specific = r_EI v_EI cos(gamma)
b = h_specific/v_inf
```

For the analytic case at 125 km altitude, `v_EI = 5.598508 km/s`. Selecting an illustrative inertial `gamma = -12°` gives `b = 7279.205 km` and a vacuum periapsis approximately `0.182 km` below the selected reference sphere. Atmospheric forces substantially alter the subsequent path. This does not establish a safe corridor: changing gamma can trade skipping, heat load, load factor, range and propellant in ways that depend on the actual vehicle.

For atmospheric calculations, use velocity relative to the rotating atmosphere and wind:

```
v_air = v_inertial - omega_Mars cross r - v_wind
q_dynamic = rho |v_air|²/2
D = q_dynamic C_D A,  L = q_dynamic C_L A
ballistic_coefficient = m/(C_D A)
```

Body attitude, angle of attack, sideslip and bank determine aerodynamic force/moment directions. A hypersonic heating correlation may have the form `q_dot = k sqrt(rho/R_n) |v_air|³`, but its coefficient and regime must be appropriate to the gas, shape and chemistry. It is not a substitute for aerothermal/TPS analysis. Current NASA high-mass Mars work examines interactions between supersonic flow, engine plumes and vehicle geometry; low-mass rover hardware should not be scaled by mass alone. NASA powered-descent wind-tunnel study, NASA high-mass EDL shortfalls.

10. Terminal guidance, orientation and physical touchdown

The local landing-frame dynamics must retain thrust direction, local gravity, drag, mass flow and attitude dynamics. For a body-to-inertial orientation `R(q)` and body-axis thrust `T_body`:

```
r_dot = v
v_dot = gravity + R(q) T_body/m + aerodynamic_force/m
m_dot = -|T_body|/(g0 Isp)
I omega_dot + omega cross (I omega) = commanded_torque + external_torque
q_dot = quaternion_kinematics(q, omega)
```

A spacecraft in cruise does not have a universal correct nose direction: thermal, power and communications requirements can set attitude. If the visualization labels an attitude as **prograde**, use the velocity relative to the selected central body, not its absolute Galactic velocity. A departure burn should show its actual thrust direction; braking usually points thrust against the relevant planet-relative motion, while final hover aligns thrust approximately against local gravity.

For the current model's nose axis `a_hat = +Y`, a shortest-arc display quaternion that aligns it to a requested unit direction `d_hat` can be constructed as `q = normalize([a_hat cross d_hat, 1 + a_hat dot d_hat])`, using vector-first/scalar-last notation. The antiparallel case requires an explicit perpendicular rotation axis; roll requires a secondary reference and continuity handling. Quaternion interpolation should avoid sign flips between equivalent `q` and `-q`. This display construction is separate from integrating physical attitude/actuator limits.

For a simple drag-free vertical braking comparison, assume downward speed `v_down = 100 m/s`, constant local gravity `g = 3.713194 m/s²`, and a constant commanded net upward acceleration of `a_net = 5 m/s²`:

```
t_brake = v_down/a_net = 20 s
h_brake = v_down²/(2 a_net) = 1000 m
T(t) = m(t) [g + a_net]
Delta-v_propulsive = (g+a_net) t_brake = 174.263881 m/s
m_ignition = m_touchdown exp(Delta-v_propulsive/[g0 Isp])
```

At an assumed `Isp = 330 s` and retained touchdown mass of `200,000 kg`, this model requires `211,064.920 kg` at ignition and consumes `11,064.920 kg` propellant. To keep the net acceleration constant as mass falls, thrust decreases from `1.839050 MN` to `1.742639 MN`. Holding thrust constant would produce a different trajectory. The 1,000 m ideal braking height contains no ignition delay, terrain, sensor, guidance, throttle or reserve margin and is not a commanded ignition altitude.

A reachable landing design must satisfy available thrust, throttle floor, gimbal/attitude bounds, propellant and control authority throughout the trajectory. A necessary vertical deceleration condition is `T_vertical/m > g` while reducing downward velocity; tilting diverts thrust from vertical support. Include horizontal divert, wind rejection, engine startup and terrain clearance in the reachable-set calculation rather than merely changing a visual pitch angle.

Touchdown success requires measured support/contact, bounded vertical and lateral speeds, acceptable attitude/rates, and stable support over a dwell period. Landing gear can be modeled initially with compliant normal forces `F_n = max(0, k penetration - c v_normal)` and friction bounded by the contact normal load. The normal/contact sign convention must be consistent. A depleted tank or impact cannot be converted into a successful landing by assigning a final position. After a terminal event, contact processing should terminate or settle under a bounded workload; it should not run an enormous fast-forward catch-up loop during the touchdown frame.

11. Starship public-data profile and what is still unknown

The public-data profile is preserved separately in starship_public_profile.json. These values are an attributed snapshot of the official design page, not a configuration-controlled vehicle interface document. SpaceX can revise them. The official page describes on-orbit refilling as part of the Mars architecture. SpaceX Starship design, checked 26 October 2026.

Public item	Attributed design-page value
Full system height / diameter	124 m / 9 m
Starship upper-stage height / diameter	52 m / 9 m
Upper-stage propellant capacity	1,600 tonnes
Upper-stage published aggregate thrust	1,614 tonne-force

Public item	Attributed design-page value
Published payload capacity	More than 100 tonnes
Upper-stage engine arrangement	Three Raptor and three Raptor Vacuum engines
Super Heavy height / diameter	72 m / 9 m
Super Heavy propellant capacity	3,650 tonnes
Super Heavy published thrust / engine count	8,240 tonne-force / 33 engines
Propellant constituents	Liquid methane and liquid oxygen

The listed aggregate thrust is not a throttle/altitude-dependent engine map, nor proof that that thrust is usable in a particular Mars landing configuration. Converting tonne-force uses `1 tf = 9,806.65 N`. The profile intentionally leaves dry mass, usable landing reserve, vacuum/ambient Isp maps, mixture ratio, throttle floor, engine-restart envelope, thrust transients, aero/TPS coefficients, landing stability limits, and certified Mars entry/landing corridor **unknown** unless a configuration-specific primary source is supplied.

The application's **EX-1** experimental transfer tug is a separate vehicle. Its configured 3,500 s transfer Isp is not a methane/oxygen Raptor specification. A complete EX-1 simulation demonstrates the application's modeled trajectory and event sequence only. It cannot validate Starship performance, predict real Starship tank sizes or serve as a universal Mars landing program. The analytic example's assumed 380/330 s Isp values also remain distinct from the public-data profile's unknown performance maps.

12. Complete Earth–Mars application run and telemetry contract

The renderer ingests the deterministic complete run supplied by the application's telemetry exporter and appends an evidence section to HTML/PDF. It reports the actual vehicle identifier and profile, epoch, achieved phases, burns, mass/fuel changes, touchdown result and any absent required field. No placeholder values are substituted for missing telemetry.

The expected attachments are earth_mars_run.json and earth_mars_run.csv, with any separate event history. The document manifest hashes those exact files. The PDF includes representative plots/tables; the full resolution records remain in the attachments. Until the exporter has provided these files, the generated evidence section explicitly says that no full application run has been ingested and makes no completion claim.

Telemetry group	Fields needed for useful reconstruction
Identity/provenance	Run/ship/profile IDs, source revision, ephemeris and force model versions, deterministic seed, units, frame and time scale
Time/events	Simulated UTC, integration seconds, phase, event identity, event time, selected/achieved clock rate
Translational state	Heliocentric and/or barycentric position and velocity, target-relative state, local position and velocity
Propulsion	Total mass, propellant remaining, thrust, throttle, mass-flow/Isp configuration, burn impulse and jettison events
Attitude	Nose/thrust vector, quaternion convention, angular rates, commanded pointing mode
Approach/contact	Altitude definition/datum, vertical/lateral speed, target range, leg loads, contact dwell and success/failure reason
Communications	Range, geometric/retarded light time distinction, RF/optical frequency, powers, gains, losses, received power and link margin

Telemetry group	Fields needed for useful reconstruction
Error/performance	Integration residuals/tolerances, navigation covariance, missed deadlines, frame-time percentiles, model exceptions

For an RF link, a basic accounting relation is `P_r,dBW = P_t,dBW + G_t,dBi + G_r,dBi - FSPL_dB - L_other,dB`, with `FSPL = 20 log10(4 pi range/lambda)`. Received power alone is not a data-rate or availability guarantee: noise temperature/bandwidth, coding, pointing, occultation, plasma and hardware constraints are additional inputs. Propagation latency and a live image feed are different data products; a simulation view must not be described as live spacecraft camera data.

For a first-order link-budget audit, distinguish carrier power, spectral noise density and the chosen data rate:

```
lambda = c/frequency
noise_density_W_per_Hz = k_B T_system
(C/N0)_dBHz = carrier_dBW - 10 log10(k_B T_system)
(Eb/N0)_dB = (C/N0)_dBHz - 10 log10(bit_rate_bits_per_s)
margin_dB = available_EbN0 - required_EbN0 - implementation_margin
first_order_one_way_Doppler_Hz = -frequency * range_rate/c
vacuum_one_way_delay = range/c
Shannon_upper_bound_bits_per_s = bandwidth * log2(1 + signal_to_noise_power_ratio)
```

The Doppler sign above makes receding positive range rate a negative frequency shift. A coherent two-way transponder needs its turnaround ratio and both light-time legs; it is not the same observable. Thermal noise uses absolute temperature in kelvin and Boltzmann's constant. Receiver noise figure, antenna efficiency/pointing, bandwidth, modulation/coding thresholds and ground-station conditions must be explicit. Shannon capacity is an ideal theoretical upper bound, not a usable guaranteed telemetry data rate. Consult versioned station/interface models rather than assuming every DSN antenna has the same gain or temperature. JPL DSN telecommunications handbook.

At interplanetary latency, the final descent needs onboard state estimation and guidance; a terrestrial viewer cannot close the touchdown control loop as if it were a local joystick. Range, Doppler and angular observations constrain different components of navigation and are not interchangeable measurements. NASA navigation fundamentals.

The run should preserve the full sequence `READY -> ASCENT -> WAITING WINDOW -> TRANSFER -> CAPTURE ORBIT -> BRAKING + DESCENT -> LANDED`, or record the actual alternative/abort sequence. Automated reporting must read event records rather than infer successful landing from the final image. Every field needs a stable definition so a change in reference frame or phase cannot masquerade as a velocity impulse.

An interactive Play/reset epoch and a deterministic reference-run epoch serve different purposes. Interactive initialization should capture the current time once for the whole shared simulation. A regression run should deliberately freeze its epoch and identify it in the exported metadata so results can be reproduced. Neither the TDB study example nor a deterministic telemetry fixture should be presented as live current-time observations.

13. Turning the study into a vehicle-specific mission design

1. Freeze vehicle configuration, payload, mass properties, available propellant after Earth-orbit refilling, reserves, and validated engine/aerodynamic models. Select launch/landing sites and terrain/atmospheric assumptions.

2. Acquire versioned ephemerides, orientation kernels and time/Earth-orientation data; validate frame conversions and units with known geometric cases.

3. Search departure/arrival grids with Lambert seeds. Retain admissible branches under launch-energy, flight-time, arrival-speed, plane, thermal, power, communications and reserve constraints.

4. Design the Earth parking/escape geometry and finite burns. Produce state-vector targets and covariance requirements rather than reusing the analytic reference's coordinate angles as commands.

5. Refine the trajectory in a consistent perturbed numerical model. Design navigation measurements and correction opportunities; propagate covariance and operational uncertainties.

6. Select capture/direct-entry architecture and target B-plane/entry states. Optimize the entry corridor, bank/angle-of-attack history, propulsion transition and reachable landing region with the actual vehicle.

7. Run atmosphere, navigation, mass, engine, aero, wind and terrain dispersions. Track joint constraints and probabilities; assess abort/divert options and required reserves.

8. Verify six-degree-of-freedom guidance, sensors and actuators, software/hardware timing, plume/ground interactions and touchdown stability. Use appropriate independent simulation, test data and flight qualification processes.

9. Export the complete trajectory, phases, state/attitude histories, mass/propellant ledger, covariance/dispersions, source versions and all failures for independent review. A successful nominal application run is one test artifact within this larger process.

14. Primary references and scope of evidence

Reference	Use in this study
NASA/JPL Horizons API	Exact boundary-state query settings and reproducible response snapshots
JPL Horizons manual	Object centers, reference frames, epochs and ephemeris interpretation
NAIF state vectors	Distinguishing geometric states from corrected observed directions
NAIF time subsystem	Time-scale definitions and conversion requirements
JPL astrodynamic parameters	AU, day and gravitational parameters
JPL planet physical parameters	Reference equatorial radii
JPL approximate elements	Nominal semimajor axis for the analytic comparison
JPL DE440/DE441 paper	Observations-based precision ephemeris modeling
NASA trajectory fundamentals	Transfer geometry context
NASA trajectory browser methodology	Efficient Lambert survey seeds and their limitations
Izzo, Revisiting Lambert's Problem	Primary algorithm research for robust Lambert families
NASA Glenn specific impulse	Propulsion definitions
NASA rocket equation	Propellant and delta-v relation
NASA Mars-GRAM 2024	Atmospheric environment/dispersions
NASA powered-descent experimental study	Aero/plume/geometry interactions in high-mass descent research
NASA high-mass EDL shortfalls, July 2024	Need for high-mass-specific entry/descent validation
SpaceX Starship design	Attributed public dimensions, capacities and architecture snapshot
JPL DSN telecommunications handbook	Versioned station, RF and navigation interface models
NASA navigation fundamentals	Range, Doppler and angular measurement roles

The worked values are calculations from the stated assumptions and retained source data. They are not figures attributed to a real NASA, SpaceX or US Navy flight plan. The public-data profile, mathematical study and experimental

application run remain separate evidence categories throughout the attachments.

15. Ingested application telemetry evidence

No complete application run has been ingested. The report currently contains mathematical reference cases only. The renderer can be rerun with the exported JSON and CSV.

16. Artifact provenance manifest

SHA-256 values identify the exact study inputs and telemetry attachments used for this export. Calculated references, public design data and measured simulation outputs have separate source files.

Input path	SHA-256
earth_mars_mission_study.md	f6de4de458cbd890e2ae58892a56c4b81d83bb1093e2238d786b0cd32a8cbe36
render_earth_mars_study.py	cb7e1cfed92f35d55454b4ea80bb7e3dd53f8943c610b438747715d6b97dbef0
earth_mars_reference.py	61fe285c5f3134bb752868d14a9a3e9c4ad1bc33d57227e6136d83a102bf65f6
earth_mars_reference_results.json	3bf917d3207c2cd79ca42ddf4e956f738a1e80f2d0a667114b30dc9dcb53578b
starship_public_profile.json	261965ea90233a9a9078af9d0d5904d8873222b02e6cbd3d7c12305f58a708b7
earth_mars_data/horizons_399.json	7a9c062738d154df75e167ad79ece79dda6ac52980f2333546447f4e67cbf8dd
earth_mars_data/horizons_399.provenance.json	1266a4b54fbb9fe5135205653510ca6298bbb148e1da2fbb079f79ef124e8df7
earth_mars_data/horizons_499.json	2908141646aadaaa7c7f7f26f7e51e91572c6324cb6458e9b549eb26e8e7d8ed
earth_mars_data/horizons_499.provenance.json	12b080e0fcd20b690bee46fa6c68dd1e74e4162c5960311f2653959771aeb846

Renderer packages: {"Markdown":"3.11.2","reportlab":"5.1.0"}.